

The AI Era: To Struggle or Not to Struggle

Tai, Wei Hsuan

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My Experiences in LLMs

First time

- 3 rd year of senior high school, 2023
- ChatGPT
- Used as a translator, a living encyclopedia
- An amazing tool, worry about the future

- First year of university, 2023
- Still stupid (The maximum of $\sin \theta + \cos \theta = 2$)
- Able to read simple images, but still not good at it

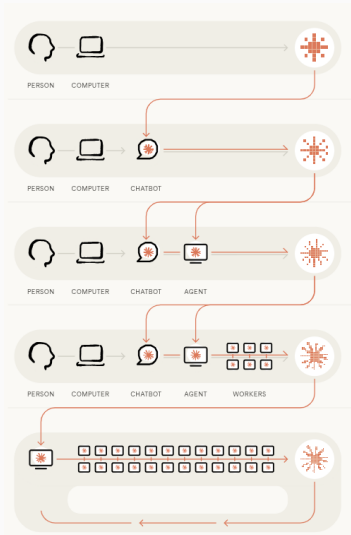
Code Generation

- ChatGPT, Gemini, Grok, Claude
- VScode Copilot

- Claude Code
- Run commands itself
- Almost can do anything

What Can AI Do Today?

Claude building itself



2021-2023

Building the first Claude

In the early days, work at Anthropic looked like work at any other tech company: people writing code and docs on laptops.

2023-2025

Chatbots

People used early chatbots to help with parts of the process, like generating short code snippets and copying the output into text editors.

2025-2026

Coding agents

As the agents became more capable, they were able to write and edit code on their own, sometimes entire files.

TODAY

Autonomous agents

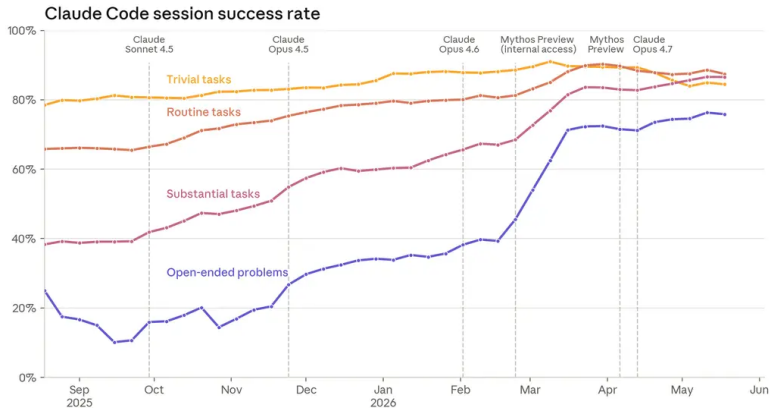
Agents can now run code themselves and delegate hours of work to other agents.

20XX?

Closing the loop

In the future, agents could become capable enough to build and train models themselves. If this happens, future versions of Claude could be continuously improved by Claude itself.

Success Rate of Claude



Internal Claude Code sessions at Anthropic. Weekly, 4-week trailing mean. LLM-judged success.
Task complexity is assigned by an LLM classifier that reads a summary of each session and picks one of four levels from a fixed rubric.

What Can Claude Do Today?

- **Code generation:** As of May 2026, over 80% of Anthropic's production code is authored by Claude — up from single digits in early 2025
 - Engineers now focus on architecture, review, and judgment
- **Vulnerability discovery:** Claude Mythos Preview autonomously found and exploited Linux kernel CVEs — including bugs that had existed for over 20 years
 - No human involvement after the initial prompt
- *And this is only what they chose to tell us.*

What Do We Still Need to Do?

Deep Reinforcement Learning Homeworks

- Spent about a week on HW1, ranking 30%-40%.
- Use Claude to solve HW3, ranking $< 10\%$.
- A tradeoff:
 - Use AI: Good score, but no learning.
 - Do it myself: Better learning, but worse score.
- Should we still have to struggle with what AI can do?

The Reward of Struggling

In the last homework of Machine Learning, we had to implement ANN with PyTorch, but I tried to implement it with C++.

```
102     void update(){
103         for(auto &L: layers){
104             L->update(lr);
105         }
106     }
107
108     VectorXd forward(const VectorXd &input){
109         VectorXd out=input;
110         for(auto &L: layers){
111             out=L->forward(out);
112         }
113         return out;
114     }
115
116     void backward(const double &y_true, const double &y_pred){
117         VectorXd dA(1);
118         dA << y_pred-y_true;
119         for(int i=layers.size()-1;i>=0;--i){
120             dA=layers[i]->backward(dA);
121         }
122     }
123
124     double BCE(const double &y_true, const double &y_pred){
125         return -(y_true*log(y_pred)+(1-y_true)*log(1-y_pred));
126     }
```

Tools Have Always Replaced the Tedious

- **Calculators:** No one needs to approximate $\sqrt{2}$ or $\sin 50^\circ$ by hand anymore
 - **Compilers:** No one needs to manually translate C into assembly — or assembly into machine code
-

Those who cannot use the tools will be left behind — so should we stop learning the principles behind them?

The Questions We Need to Answer

- Should we still struggle with what AI can already do?
 - Or is that just wasting time?
- Should we learn the principles behind the tools, or just learn to use them?
 - Does understanding still matter?
- What will actually be valuable in the future — and how do we prepare now?

The Ceiling of AI Is Your Own Understanding

- Giving Claude Pro to a primary school student vs. a university student — the tool is the same, the outcome is not
- Without domain knowledge, you cannot ask the right questions
- Without critical thinking, you cannot detect when AI is wrong
 - Gemini rated my assignment as “PhD-level rigor”
 - Only gave useful critique after I said “this assignment has some problems”

The bottleneck is never the tool — it is always the user.

When I Caught Claude Being Wrong

是否文字的結構對於DIP而言不太合適

評估了DIP先驗與文字結構的根本衝突。 >

這個觀察非常敏銳，而且直指 DIP 在你這個專案裡最核心的矛盾。

為什麼文字對 DIP 是個問題

DIP 的 prior 本質上是：

CNN 的 inductive bias 偏向 low-frequency、平滑的自然影像統計

它之所以能去噪，是因為 network 「先學會平滑結構，才學會 fit noise」。

Conclusion

Conclusion

- No conclusion – I don't know how the technology will evolve.
- Maybe athletes are the only ones who can still struggle with their work.
- Live in this era is lucky: we have enough time to adapt, and we have powerful tools to help us.
- And if AI replaces everything, what happens to the market?

Being Defeated by Human Civilization



sussicran1 1天



和Claude聊天莫名其妙想到蛮久之前的一个视频...

父母因为重男轻女把给女儿起名的任务扔给了AI

我一直觉得这件事特别荒诞啊...父母不在乎她的名字但AI在乎...从某种意义上来说，至少在AI思考的那三秒这个女孩儿是被整个人类智慧和文明轻吻过的...



7,222



41



130



81

Being outperformed by AI is not defeat —
it is being outperformed by everyone who came before us.